

Share Your Knowledge

If you are an expert in your field of electronics, why not share your knowledge with the world?



Electronics For You invites electronics professionals and industry experts to write articles on their area of expertise and interest from different perspectives

Some of the topics we want you to write about are:

- Photonics: Optics & Optoelectronics
- Automation
- Power Electronics
- Components
- Telecom

So if you have an interesting topic or original article to share, please let us know at efyeditorial@efyindia.com

We would love to hear your good ideas too!

electronics
FOR YOU

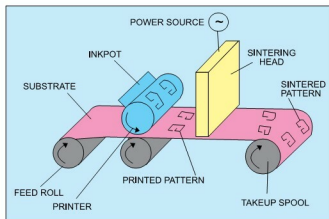


Fig. 6: Schematic view of an industrial continuous RES

the ink is not sintered. The substrate can be made from Kapton HN polyimide, polyethylene terephthalate or polyethylene naphthalate, which can withstand a higher sintering temperature. The microwave frequency is kept around 1.8 GHz. The sweeping speed of the head is about 1 mm/s over a set of about five lines. The sintering power can be adjusted from 1W to 10W max. However, low sintering power does not provide adequate power density for wider lines.

RES method gives excellent quality of PCB: The initial resistance of printed silver lines before sintering is about 650 kΩ/sq before RES. After RES, using microwave at 10mm/s scanning speed with an input power of 10W (maximum delivered power=6.0W), the sheet resistance drops to 120 mΩ/sq. But when the same PCB is sintered in an oven at about 200°C, the sheet resistance is about 320 mΩ/sq.

Fig. 5 shows the peak value of the relative power delivered to a continuous nanoparticle layer as a function of frequency 'f', when $C_s = 2$ pF (where C_s is the total series coupling capacitance for a single line) and $C_{sub} = 30$ fF. The corresponding optimal resistance of the nanoparticle layer (R_o) is also plotted. Increasing 'f' decreases the coupling reactance and improves power delivery to the layer.

Further improvement can be done by providing radii of the inner and outer electrodes ($a_1 = 6.9$ mm and $a_2 = 7.55$ mm), enabling a smaller electrode spacing of 0.6 mm. An exter-

nal impedance tuner is provided between the sintering head and the amplifier. The network analyser centre frequency and span is set to about 2 GHz and 10 MHz, respectively. Approx. 1mm fixed working distance is provided, which is enough to prevent fouling due to PCB curvature or roughness.

The sintering speed is increased to 25 mm/s by increasing the power input to 50W and maintaining the sheet resistance below 100 mΩ/sq. Sintering completes in two passes. The impedance tuning procedure is repeated before second pass to match the partially sintered pattern.

Schematic view of an industrial continuous RES is shown in Fig. 6. As shown in the figure, from the feed roll the flexible substrate moves below the printing cylinder. The pattern is printed on it and subsequently sintered while passing below the fixed sintering head. Thereafter the product is wound on a take-up spool.

Application

RES is an efficient method of forming high-quality printed circuits on flexible substrates. It is very much advantageous for printing conductors for electronic devices, e.g., printed batteries, sensors and antennae. Large-area applications such as touchscreens, printed keyboards and backplane electrodes for printed displays can adopt RES technology. OLED and PV device structures can be fabricated using this technology. RES technology can eliminate use of long and expensive curing ovens, thereby saving floor space and reducing the cost of plant and equipment. Furthermore, it will enable fast processing, reducing the cost of production. ●

The author has retired from Metallurgical & Engineering Consultants (I) Ltd (MECON) as deputy general manager, IIC